Project Report

Income Classification & Customer Segmentation
US Census Bureau Data (1994–1995 CPS)

1. Overview

The goal here was to build two models on census data (~200k records, 40 features from the 1994/1995 CPS) for a retail marketing client:

- A classifier that predicts whether someone makes over or under $50k
- A segmentation model that groups people into distinct marketing audiences

2. Data

The dataset has 42 columns total (40 features + instance weight + income label). Mix of categorical and numerical. Lots of missing values encoded as ? which we handle during preprocessing.

High Level Details:

- ~200k rows
- Heavy class imbalance on the label -- way more people in the <50k bucket
- Different categories of features: Demographics (age, race, sex), Employment details (occupation, industry, weeks worked), Financial information (capital gains/losses, dividends).

3. Preprocessing

For both models:

- I've Replaced ? with NaN, then filled categorical NaNs with 'Unknown' and numeric NaNs with 0
- One-hot encoded all categorical features using pd.get_dummies
- Scaled everything with StandardScaler
- Dropped weight column since it's for sampling purposes, not really a feature

4. Classification

Approach

Used a Random Forest with 100 trees. Went with class_weight='balanced' since the >50k class is way underrepresented (~6%). Did an 80/20 stratified split.

I considered logistic regression too but RF generally handles the mix of encoded categoricals + numerics better without needing as much feature engineering. Also gives us feature importances for free which the client wanted.

Results

	precision	recall	f1-score	support
0	0.96	0.99	0.98	37429
1	0.73	0.38	0.50	2476
accuracy			0.95	39905
macro avg	0.84	0.68	0.74	39905
weighted avg	0.95	0.95	0.95	39905

ROC-AUC: 0.9391

The model is solid at ranking people by income likelihood (AUC ~0.94). The precision on the >50k class is decent (73%) but recall is low (38%). This means if we flag someone as high-income we're usually right, but we're missing a lot of actual high earners. For marketing purposes this is probably fine, better to target a smaller confident group than waste budget on bad leads.

Top Features

The biggest predictors were weeks_worked_in_year, capital_gains, age, and education. People who work more weeks, have investment income, are older (more experienced), and have higher education tend to earn more.

5. Segmentation

Approach

Used MiniBatch KMeans with k=4 clusters. Tried a few values of k (3, 4, 5) and 4 gave the most interpretable groupings. Used MiniBatch variant because regular KMeans was pretty slow on 200k rows with all the one-hot encoded features.

Used TruncatedSVD to project down to 2D for the scatter plot visualization since PCA needs dense matrices and our encoded data is huge.

Results

Found 4 segments:

Cluster	Avg Age	Avg Weeks Worked	Dominant Education	Interpretation
0	40	46	HS graduate	Full-time workers / blue collar
1	9	0	Children	Dependents / minors
2	46	18	5th-6th grade	Part-time / low education
3	46	30	HS graduate	Mixed employment / older adults

Marketing Recommendations

- **Cluster 0 (Working adults):** Bread-and-butter consumers. Employed, working class. Target with everyday retail, practical goods, promotions.
- **Cluster 1 (Kids):** Not directly targetable but represents families. Market family/kids products to their households.
- **Cluster 2 (Part-time / lower education):** Budget-conscious segment. Discount offers, value products.
- **Cluster 3 (Older mixed employment):** Could include semi-retired, part-time workers. Health products, leisure, home goods.

6. What I'd do with more time

- Try XGBoost or LightGBM for the classifier, probably would boost recall on the minority class
- Do proper hyperparameter tuning with cross-validation (GridSearchCV or RandomizedSearch)

- Try DBSCAN or hierarchical clustering for segmentation to see if non-spherical clusters emerge
- Build a proper elbow plot / silhouette analysis to pick optimal k more rigorously
- Look into SMOTE or other oversampling techniques for the class imbalance

7. References

- Scikit-learn docs: https://scikit-learn.org/
- Pandas docs: https://pandas.pydata.org/
- Original dataset: US Census Bureau, Current Population Survey 1994-1995